

Camila VILLALBA HEUER

Java Backend Developer | SQL & Relational Databases



PROFESSIONAL SUMMARY

Java Backend Developer (Junior) with an engineering background and hands-on experience gained through academic and self-directed projects. Focused on building backend systems with Java, Spring Boot, SQL, and microservices architecture, applying clear validation rules, explicit state management, and well-documented REST APIs, with experience implementing Spring Security and JWT-based authentication, along with backend testing practices. Background in safety-critical environments (nuclear engineering), bringing a systems-thinking mindset oriented to reliability, data consistency, and controlled processes. Motivated to continue growing as a backend developer through real-world projects and collaborative teams.

CONTACT

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PERSONAL INFORMATION

Age: **30 years old**

Nationality: **Argentine**

LANGUAGES

Spanish (Native)

English (B1)

TECHNICAL SKILLS

- Backend: Java (JDK 17+), Spring Boot, REST APIs, Microservices (Spring Cloud), Spring MVC, JPA/Hibernate, Java EE, layered architecture, DTO pattern, Functional Programming.
- Security: Spring Security, JWT authentication, role-based authorization, service-to-service security, OAuth2 fundamentals
- Testing & Quality: JUnit 5, Mockito, MockMvc, Bean Validation, API testing (Postman), manual testing, CI/CD (GitHub Actions)
- Databases: MySQL, SQL, relational modeling, joins, subqueries, transactions, constraints, indexes, stored procedures.
- Frontend: React (basic), HTML5, CSS3, JavaScript
- Tools & DevOps: Docker, Maven, Git/GitHub, Postman, Swagger/OpenAPI
- Methodologies: Agile (Scrum fundamentals), Clean Code practices
- Additional: Python (Pandas, NumPy), Power BI, Power Query

PROFESSIONAL EXPERIENCE

August 2025 – Present

BACKEND DEVELOPER — INDEPENDENT PROJECTS

- ◇ Development of REST APIs using Java 17 and Spring Boot following MVC architecture and layered design principles.
- ◇ Design and implementation of microservices architecture using Spring Cloud (Eureka, API Gateway, Config Server).
- ◇ Data persistence with Spring Data JPA, Hibernate and relational databases.
- ◇ Implementation of validation, global exception handling and secure API design.
- ◇ Unit and integration testing using JUnit and MockMvc.
- ◇ Containerization with Docker and version control using Git/GitHub.
- ◇ Application of Agile methodologies in project development.

March 2023 – October 2025



HEAD OF MAINTENANCE & OPERATIONS at *RA-10 Nuclear Reactor*
– CNEA

- ◇ Development and optimization of operational and maintenance procedures for a nuclear research reactor.
- ◇ Technical analysis, documentation management and regulatory compliance.
- ◇ Cross-disciplinary collaboration in high-reliability and safety-critical environments with strict quality and traceability requirements.
- ◇ Process analysis, troubleshooting and continuous improvement initiatives.
- ◇ Led Safety Culture initiatives within the RA-10 project, promoting risk anticipation, responsible decision-making and interdisciplinary collaboration.
- ◇ Designed technical training and knowledge-transfer activities for operators and maintenance personnel, improving system understanding and operational readiness.

Engineering background applied to software development through analytical problem-solving, structured system design and strong quality and safety practices.

ADDITIONAL PROJECTS

REST API - Sales & Inventory Management System

Spring Boot · JPA · MySQL / H2

- Designed and implemented a RESTful API applying MVC architecture, DTO pattern, and business validations.
- Developed full CRUD operations with relational persistence and transactional consistency.
- Applied layered architecture and clean separation between controllers, services, and repositories.

Supermarket Management System

Java · functional programming

- Implemented a backend-focused Java application applying functional programming concepts (streams, lambdas, method references)
- Developed business logic for data processing, filtering, aggregation, and transformation.

Python Mini-Projects Collection

Python · Testing · CI

- Developed multiple backend-oriented Python applications, including an inventory management system with dual persistence (in-memory / MySQL).
- Implemented automated testing (90+ tests) and CI pipelines using GitHub Actions.
- Applied modular design principles to support scalability and maintainability.

Relational Database Design Projects

MySQL · SQL

- Designed and implemented relational database schemas for e-commerce, bookshop, and clinic systems.
- Wrote advanced SQL queries using joins, subqueries, aggregations, and transactions.
- Implemented stored procedures and indexes to encapsulate business logic and improve performance.

SOFT SKILLS

- Analytical thinking
- Problem-solving
- Attention to detail
- Fast learning & adaptability
- Clear communication
- Methodical & organized
- Team collaboration

SELECTED SOFTWARE PROJECTS

January 2026

Microservices Architecture - Online Appliance Store (Spring Boot + Spring Cloud + Docker)

Designed and implemented a distributed e-commerce system based on microservices architecture, applying Spring Cloud patterns to ensure scalability, fault tolerance, and data consistency.

- ◊ Built a microservices ecosystem with Eureka (Service Discovery), API Gateway, Config Server, and OpenFeign for inter-service communication.
- ◊ Implemented fault tolerance with Resilience4j and enforced explicit state management for carts (CREATED, SOLD) and products (ACTIVE, DISCONTINUED) to prevent inconsistent operations.
- ◊ Applied Bean Validation on DTOs, global exception handling, and documented REST APIs with Swagger (OpenAPI).
- ◊ Implemented distributed security with Spring Security and JWT, separating user authentication/authorization from service-to-service security through a dedicated technical service identity and role-based access control.
- ◊ Applied controlled state transitions inspired by safety-critical engineering systems, ensuring consistency without tight inter-service coupling.
- ◊ Refactored DTOs to Java records and separated request/response models to improve code clarity and maintainability.

October 2025

Java Desktop Application – Grooming Manager (Swing + JPA + MySQL)

Designed and developed a Java-based desktop management system applying layered architecture principles and strong data consistency rules.

- ◊ Built a Java Swing application with MVC-style layered architecture (UI, business logic, persistence).
- ◊ Modeled domain entities with complex relationships and implemented advanced data validation to prevent duplicated or inconsistent records.
- ◊ Designed persistence layer using JPA (EclipseLink) with MySQL, managing ORM mappings and transactional operations.
- ◊ Managed the project with Maven and implemented usability improvements such as autocomplete for data entry.

EDUCATION

February 2022 – December 2022



SPECIALIZATION IN TECHNOLOGICAL APPLICATIONS OF NUCLEAR

ENERGY at *Balseiro Institute and Faculty of Engineering – University of Buenos Aires (UBA)*

February 2015 – September 2021



CHEMICAL ENGINEERING – SPECIALIZATION IN ENVIRONMENTAL ENGINEERING at *Faculty of Exact, Chemical and Natural Sciences – National University of Misiones*

RELEVANT INTERESTS

Interested in backend architecture, REST API design, and the Java/Spring Boot ecosystem. Particularly motivated by relational database design, SQL optimization, ORM usage (JPA/Hibernate), and improving system reliability through explicit state management and clean data modeling. Interested in practices that enhance code quality and maintainability, such as clean code principles, testing, and version control workflows.

ADDITIONAL TRAINING

- ◇ Backend Development (TodoCode Academy): Microservices with Spring Cloud, API Development with Spring Boot, Java Web (JSP, Servlets, JPA), Functional Programming, Object-Oriented Programming, Software Testing and Quality, Security in Java with Spring Security
- ◇ Databases: SQL (MySQL) – Self-taught (SQLBolt, DB Fiddle, XAMPP + phpMyAdmin)
- ◇ Python: Introduction to Python, Python for Data Analysis (Pandas, NumPy), Testing & CI/CD with Python (unittest, GitHub Actions)
- ◇ Web Basics: HTML, CSS, JavaScript
- ◇ Data & Analytics: Power BI and Power Query (Santander Academy), Introduction to Data Analysis (Unicorn Academy)